

Case 2: $x_1 = 0$ with equation (17.18) implies $x_2 = \frac{I}{p_2}$. Since this is positive, $\lambda_2 =$

0, so equation (17.20) implies $\lambda_3 = \frac{1}{\left(1 + \left(\frac{I}{p_2}\right)^2\right)} p_2$.

$\lambda_1 = \lambda_3 p_1 - 1$ (from $x_1 = 0$ and equation (17.19)). $\lambda_1 = \frac{p_1 p_2}{(p_2 + I)^2} - 1$. For this to

be greater than zero it is required that $\frac{p_1 p_2}{(p_2 + I)^2} \geq 1$, that is $I \leq (p_1 p_2)^{\frac{1}{2}} - p_2$.

Utility equals $\frac{x_2}{(1 + x_2)} = \frac{I}{(p_2 + I)}$.

$$(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(0, \frac{I}{p_2}, \frac{p_1 p_2}{(p_2 + I)^2} - 1, 0, \frac{p_2}{(p_2 + I)^2}\right).$$

Case 3: By symmetry, the solution is

$$(x_1^*, x_2^*, \lambda_1^*, \lambda_2^*, \lambda_3^*) = \left(\frac{I}{p_1}, 0, 0, \frac{p_1 p_2}{(p_1 + I)^2} - 1, \frac{p_1}{(p_1 + I)^2}\right)$$

and for this case to hold it is necessary that $\frac{p_1 p_2}{(p_1 + I)^2} \geq 1$, or $I \leq (p_1 p_2)^{\frac{1}{2}} - p_1$.

To summarise, suppose $p_1 = p_2 = p$, then $(p_1 p_2)^{\frac{1}{2}} - p_1 = (p_1 p_2)^{\frac{1}{2}} - p_2 = 0$. So since $I > 0$, we are in the regime of Case I, and $x_1 = x_2 = (I/2p)$ at the maximum. Suppose on the other hand that $p_1 < p_2$ (the contrary can be worked out similarly). Then, $p_2 > (p_1 p_2)^{\frac{1}{2}} > p_1$. Thus, considering $I > (p_1 p_2)^{\frac{1}{2}} - p_1$, we use Case (1), or in case of $I \leq (p_1 p_2)^{\frac{1}{2}} - p_1$ we use Case (3). Case (2), that in which a positive amount of good 2 and zero amount of good 1 is consumed at the maximum, does not apply.

17.6 LET US SUM UP

In this unit we emphasized on two important tools, which have enormous applications in economics viz., unconstrained and constrained optimisation. Unconstrained optimisation problem deals with optimisation (i.e., either minimisation or, maximisation) problems where the domain of the objective function is not limited by any constraint. We emphasised the basic theory of constrained optimisation, with its division into two main parts viz., constrained optimisation in the case of equality constraints and constrained optimisation in the case of inequality constraints. There are many real-life economic problems for which unconstrained and constrained optimisation techniques are necessary. In the case of optimisation under equality constraints, we apply the Lagrangian method and in the case of optimisation under inequality constraints, we apply the Kuhn-Tucker theory. We also worked out some simple economic problems which use these techniques of unconstrained and constrained optimization.

17.7 KEY WORDS

Necessary and Sufficient Condition for Optimum : Given a function $z = f(x_1, x_2, \dots, x_n)$, the conditions are obtained as the following:

FOC (Necessary condition) for Maximum
 $f_1 = f_2 = \dots = f_n$

FOC (Necessary condition) for Minimum
 $f_1 = f_2 = \dots = f_n$

SOC (Sufficient condition) for Maximum
 $(-1)^i |H_i| > 0 \forall i = 1, \dots, n$

SOC (Sufficient condition) for Minimum $|H_i| > 0 \forall i = 1, \dots, n$

Stationary Point : In calculus, a **stationary point** is referred to a point on a graph where the tangent to the graph is parallel to the x -axis or, equivalently, where the derivative of the function equals zero.

Theorem of Lagrange : Let $f: R^n \rightarrow R$ and $g^1: R^n \rightarrow R, g^2: R^n \rightarrow R, \dots, g^k: R^n \rightarrow R$, $k < n$ be once continuously differentiable functions. Suppose x^* is a Max or a Min off on the set $S = U \cap \{x \in R^n \mid g^i(x) = 0, i = 1, \dots, k\}$, for some open set $U \subseteq R^n$. Then there exist real numbers $\mu^*, \lambda_1^*, \dots, \lambda_k^*$, not all zero, such that

$$\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = \theta_{1 \times n}.$$

Moreover, if $\text{rank}(Dg(x^*)) = k$, then we may put $\mu = 1$.

Theorem of Second Order Optimum : Suppose there exists $(x_{n \times 1}^*, \lambda_{k \times 1}^*)$ such that
 $\text{Rank}(Dg(x^*)) = k$ and $Df(x^*) + \sum_{i=1}^k Dg^i(x^*) = \theta$.

i) A necessary condition: If f has a local maximum (respectively for local minimum) on S at point x^* , then $x^T D^2 L(x^*, \lambda^*) x \leq 0$, (respectively for ≥ 0) for all $x \in N(Dg(x^*))$.

ii) A sufficient condition: If $x^T D^2 L(x^*, \lambda^*) x < 0$, (respectively for > 0), for all $x \in N(Dg(x^*))$, $x \neq 0$, then x^* is a strict local maximum (respectively for strict local minimum) of f on S .

The Kuhn-Tucker (KT) Theorem : Let $f: R^n \rightarrow R$, and $g^i: R^n \rightarrow R$, $i = 1, \dots, k$ be once continuously differentiable functions. Suppose x^* is a maximum of f on the set $S = U \cap \{x \in R^n \mid g^i(x) \geq 0, i = 1, \dots, k\}$, for some open set

$U \subseteq R^n$. Then there exist real numbers $\mu, \lambda_1^*, \dots, \lambda_k^*$,
not all zero such that
 $\mu Df(x^*) + \sum_{i=1}^k \lambda_i^* Dg^i(x^*) = 0_{1 \times n}$.

Moreover, if $g^i(x^*) > 0$ for some i , then $\lambda_i^* = 0$. If in addition, $\text{Rank} Dg_\Phi(x^*) = 1$, then we may take μ to be equal to 1. Furthermore, $\lambda_i^* \leq 0, i = 1, \dots, k$, and $\lambda_i^* > 0$ for some i implies $g^i(x^*) = 0$.

17.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See the example given in Sub-section 17.2.2.
- 2) The objective function is $y = x^2 - 6$. Therefore, from the first order condition for optimisation, we get $2x = 0$. Solving this equation, we have $x = 0$. The second order condition gives that $d^2y = 2 > 0$, which is not a function of x . Therefore, we have $d^2y = 2$ at $x = 0$. So, at $x = 0$, we reach the minima of the objective function.
- 3) The objective function is $y = x^3 - 2x^2 + x - 6$. Therefore, we get the first order condition $3x^2 - 4x + 1 = 0$ or, either $x = \frac{1}{3}$ or, $x = 1$. Note that $d^2y = 6x - 4$. So, $d^2y > 0$ at $x = 1$ and $d^2y < 0$ at $x = \frac{1}{3}$. Therefore, at $x = 1$, we have the minima and at $x = \frac{1}{3}$ we have the maxima.
- 4) The objective function is $z = x^2 - 2x - y^2$. The necessary first order condition yields $2x - 2 = 0$ and $2y = 0$. From these two first order conditions we get $x = 1$ and $y = 0$. The corresponding Hessian matrix is $|H| = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$, so we get that $|H_1| = 2 > 0$ and $|H_2| = -4 < 0$. Therefore, we have the minimum value of z at $x = 1$ and $y = 0$.

Check Your Progress 2

- 1) The Lagrange function of the problem: $L(x, \lambda) = x^2y + \lambda [200 - 2x - y]$
The FOC for this problem are:
 $2xy = 2\lambda$ -(1), $x^2 = \lambda$ -(2) and $200 = 2x + y$ -(3). Solving these three equations, we get $x^* = (200/3)$, $y^* = (200/3)$ and $\lambda^* = (200/3)^2$.
Note that CQ satisfied because $p_x > 0$ and $p_y > 0$.
The second order condition of this problem is –

$$|\overline{H}| = \begin{vmatrix} 0 & -2 & -1 \\ -2 & 2(200/3) & 2(200/3) \\ -1 & 2(200/3) & 0 \end{vmatrix}. \text{ One can check that this matrix is}$$

negative definite. Therefore, second order condition fulfilled.

- 2) The problem can be solved exactly the same way as done in the previous one.

Check Your Progress 3

- 1) See the method of solving optimisation problem under inequality constraint in Section 6.3. You should be able to get $x^* = 25$ and $y^* = (50/3)$.

Note that Lagrangian function is

$$L(x, y, \lambda) = x^{0.5}y^{0.5} + \lambda [100 - 2x - 3y]$$

The first order Kuhn-Tucker conditions are:

$$0.5x^{-0.5}y^{0.5} - 2\lambda \leq 0, x \geq 0, x[0.5x^{-0.5}y^{0.5} - 2\lambda] = 0$$

$$0.5x^{0.5}y^{-0.5} - 3\lambda \leq 0, y \geq 0, y[0.5x^{0.5}y^{-0.5} - 3\lambda] = 0$$

$$100 - 2x - 3y \leq 0, \lambda \geq 0, \lambda[100 - 2x - 3y] = 0$$

- 2) See the method of solving optimisation problem under inequality constraint in Section 17.4. You should be able to get $x^* = 6.1$ and $y^* = 6.5$.

Note that the Lagrangian function is

$$L(x, y, \lambda) = (x - 3)(y - 2) + \lambda [50 - 5x - 3y]$$

The first order Kuhn-Tucker conditions are:

$$(y - 2) - 5\lambda \leq 0, x \geq 0, x[(y - 2) - 5\lambda] = 0$$

$$(x - 3) - 3\lambda \leq 0, y \geq 0, y[(x - 3) - 3\lambda] = 0$$

$$[50 - 5x - 3y] \geq 0, \lambda \geq 0, \lambda[50 - 5x - 3y] = 0$$

17.9 EXERCISES

Q1. $f(x, y) = 2xy + 2x - x^2 - 2y^2$, find the optimum of $f(x, y)$.

Ans. $\partial f / \partial x = 2y + 2 - 2x$,

$$\partial f / \partial y = 2x - 4y.$$

$$\text{Let } \partial f / \partial x = 0, -2x + 2y = -2.$$

$$\text{Let } \partial f / \partial y = 0, 2x - 4y = 0.$$

Then $x^* = 2$ and $y^* = 1$, i.e., $(2, 1)$ is a critical point.

$$\partial^2 f / \partial x^2 = \partial / \partial x (2y + 2 - 2x) = -2$$

$$\partial^2 f / \partial y^2 = \partial / \partial y (2x - 4y) = -4$$

$$\partial^2 f / \partial x \partial y = \partial / \partial x (2x - 4y) = 2, \text{ or}$$

$$\partial^2 f / \partial y \partial x = \partial / \partial y (2y + 2 - 2x) = 2$$

$$|H| = \partial^2 f / \partial x^2 \cdot \partial^2 f / \partial y^2 - \partial^2 f / \partial x \partial y \cdot \partial^2 f / \partial y \partial x = (-2) \times (-4) - 2^2 = 4 > 0$$

$$\partial^2 f / \partial x^2 < 0. (x^*, y^*) = (2, 1) \text{ is a local maximum.}$$

Q2. Maximise utility $u = f(x, y) = xy$ subject to the constraint $g(x, y) = x + 4y = 240$. Here the price of per unit $x = \text{Rs. } 1$ and price of per unit y is $\text{Rs } 4$. The total income is 240.

Ans. Here the optimization problem is:

Objective Function: $u = f(x, y) = xy$

Subject to the constraint: $g(x, y) = x + 4y = 240$.

Step 1: Find out the slope of the indifference curve through the objective function. $-(y/x)$

Step 2: Find out the slope of the budget line i.e., the constraint. $-(1/4)$

Step 3: Equate both the slopes (Utility maximization requires both the slopes to be equal) to get $x = 4y$

Step 4: From step 3 use the relation between x and y in the constraint function to get the optimal/critical values of x & y .

$y = 30$ and hence $x = 120$.

Q3. Suppose a consumer consumes two goods, x and y and has utility function $u(x, y) = xy$. He has a budget of $\text{Rs. } 400$. The price of x is $P_x = 10$ and the price of y is $P_y = 20$. Find his optimal consumption bundle using the Lagrange method.

Ans. Here the optimization problem is:

Objective function: maximize $u(x, y) = xy$

Subject to the constraint: $g(x, y) = 10x + 20y = 400$.

This is a problem of constrained optimization.

Form the Lagrange function:

$$L(x, y, \mu) \equiv f(x, y) - \mu(g(x, y) - k)$$

$$L(x, y, \mu) \equiv xy - \mu(10x + 20y - 400)$$

Set each first order partial derivative equal to zero:

$$\partial L / \partial x = y - 10\mu = 0 \quad (1)$$

$$\partial L / \partial y = x - 20\mu = 0 \quad (2)$$

$$\partial L / \partial \mu = -(10x + 20y - 400) = 0 \quad (3)$$

From equations (1) and (2) we find:

$$x = 2y$$

Use $x = 2y$ in equation (3) to get: $10x + 20y = 400$

$$40y = 400$$

$$Y = 10$$

$$X = 2Y = 20$$



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UNIT 18 ADVANCED TOPICS IN OPTIMISATION– I

Structure

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18.0 OBJECTIVES

After going through this unit, you should be able to:

- explain the basic optimisation problems with convexity as well as concavity;
- describe whether the boundary is binding or non- binding while solving optimisation problems with inequality constraints;
- describe the constrained optimisation and inequality constrained optimisation; and
- state the complementary slackness condition and Kuhn Tucker formulations.

18.1 INTRODUCTION

Concave functions and homogeneous functions are widely observed in economic models. Concave functions serve as spending functions and homogeneous functions as demand functions. Cost and profit functions are inherently homogeneous and concave. Hence, this unit discusses the concepts of concavity and convexity in detail, including quasi-convexity and quasi-concavity. Complementary slackness conditions and Kuhn–Tucker conditions for the general mathematical programming problem, including equality and inequality constraints are also explained as well as non-negative and free variables are provided along with some applications.

18.2 CONVEXITY AND CONCAVITY

In this section we will learn about the basic concepts of convexity and concavity first and then proceed to differentiable convex, quasi/convex and concave, quasi/concave functions.

18.2.1 Convex and Concave Functions and Generalization

There are numerous unique and significant characteristics of concave and convex functions. Concave and convex functions are important subjects that are introduced here along with some of their characteristics. For optimality issues involving convex and concave functions, these qualities can be used to create adequate optimality criteria and computational frameworks. Convexity is a key idea in economics. Convexity is frequently assumed in economic models for the budget set of agents. In many cases, we suppose that agent preferences are convex and utility functions are (quasi)concave. These facts provide us with motivation to examine the (quasi)concave and (quasi)convex functions.

Convex functions are described algebraically as follows: a map f is convex iff $f(\theta u + (1 - \theta)v) \leq \theta f(u) + (1 - \theta)f(v)$, for any real numbers u, v, θ and for all $0 \leq \theta \leq 1$. In the above expression, if the inequality is a strict one, the function is strictly convex. For convex functions, a general property is that the sum of convex functions is also a convex function. Also, if a convex function is multiplied by a positive constant, we obtain another convex function.

In the case of concave map (function), the set below the function must be convex. Also, the inequality in the expression above is reversed in the case of a concave function. We will learn more about it in subsequent sections.

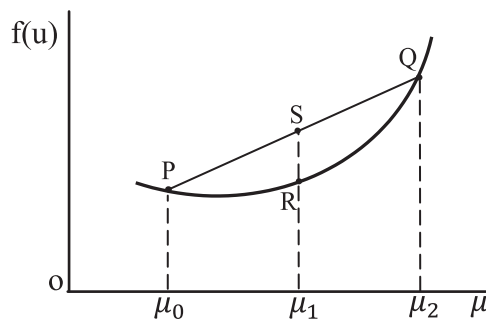


Fig. 18.1: Convex Function $v = f(u)$

Consider Fig.18.1. For mapping $v = f(u)$, let f be a differentiable convex function, and suppose $u < v$ and $0 \leq \theta \leq 1$, so that $u_0 = \theta u + (1 - \theta)v$.

Now, let points P, Q, R and S in the uv -plane be defined as shown in Fig. 18.1. Since f is convex, R cannot lie above S, therefore the slope of the chord PR cannot be more than the slope of the chord PQ. This inequality holds even as θ tends to 1, in which case R approaches P along the curve and the slope of PR approaches the tangent or slope of the curve at P. Hence the slope (tangent) of the curve at P will be less than the slope of the chord PQ. Thus,

$$f'(u) \leq \frac{f(v) - f(u)}{v - u}.$$

Similarly, as θ tends to 0, we obtain

$$f'(v) \geq \frac{f(v) - f(u)}{v - u}$$

Convex function: Assume $S \subseteq R^n$ be a convex set and $f : S \rightarrow R$. Then f is called a convex function if for all $u, v \in S$ and for all $0 \leq \theta \leq 1$, we have

$$f(\theta u + (1 - \theta)v) \leq \theta f(u) + (1 - \theta)f(v) \quad \dots(18.1)$$

The map $f : S \rightarrow R$ is called **strictly convex** on S if the above inequality is true as a strict inequality for each and, $u, v \in S$ and $\theta \in [0, 1]$.

Concave function: Let $S \subseteq R^n$ be a convex set and $f : S \rightarrow R$. Then f is called a concave function if for all $u, v \in S$ and for all $0 \leq \theta \leq 1$, we have

$$f(\theta u + (1 - \theta)v) \geq \theta f(u) + (1 - \theta)f(v).$$

The function $f : S \rightarrow R$ is called **strictly convex** on S if the above inequality is true as a strict inequality for each and, $u, v \in S$ and $\theta \in [0, 1]$. Also, $f(u)$ is said to be concave on $[a, b]$ iff the function $-f(u)$ is convex on $[a, b]$.

Now we will express the geometric interpretation of convex and concave functions. Let u and v be two distinct points in the domain of f , and consider the point $\theta u + (1 - \theta)v$ with $\theta \in [0, 1]$. So for a convexity f , the value

of f at points on the line segment $\theta u + (1-\theta)v$, is less than or equal to the height of the chord joining the points $[u, f(u)]$ and $[v, f(v)]$. For the concave function, the chord is below the function itself. Hence, a function is both convex and concave if and only if it is affine.

18.2.2 Examples of Convex and Concave Functions

Examples of Convex Functions:

- 1) Any function of the form $f(u) = au + b : a, b \in \mathbb{R}$.
- 2) Powers: $f(u) = u^p ; p \geq 1$.
- 3) Powers of absolute value: $f(u) = |u|^p$ on $\mathbb{R}$, for $p \geq 1$.
- 4) Exponential: $f(u) = e^{au}$, for any $a \in \mathbb{R}$.
- 5) Every linear transformation taking values in $\mathbb{R}$ is convex.
- 6) $f(u_1, u_2) = 2u_1^2 + u_2^2 - 2u_1u_2$.

Examples of Concave Functions:

- 1) $f(u) = -u^2$.
- 2) $f(u) = \sqrt{u}$.
- 3) Any linear function $f(u) = au + b : a, b \in \mathbb{R}$.
- 4) The map $f(u) = \sin(u)$ is concave in $[0, \pi]$.

We give below some particularly important instances of convex functions that arise very often in practice.

- 1) Suppose $f_1, f_2, \dots, f_n : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex maps. Then,
 - a) $f(u) = \sum_{j=1}^k c_j f_j(u)$, where $c_j > 0$ for $j = 1, 2, 3, \dots, k$ is a convex map.
 - b) $f(u) = \text{Max}\{f_1(u), f_2(u), \dots, f_k(u)\}$ is a convex map.
- 2) Assume that function (map) $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is a concave. Let $S = \{x : g(x) > 0\}$, and define $f : S \rightarrow \mathbb{R}^n$ as $f(x) = \frac{1}{g(x)}$. Then, function f is also convex over S .
- 3) Consider map $g : \mathbb{R} \rightarrow \mathbb{R}$ be a non-decreasing, convex, univariate and let function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex. Then, the composition function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined as $f(u) = g[h(u)]$ is a convex.
- 4) Let function $g : \mathbb{R}^m \rightarrow \mathbb{R}$ be a convex, and let function $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be an affine of the form $h(u) = Au + b$, where A is a $m \times n$ matrix and b is a $m \times 1$ vector. Then, the composite function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined as $f(u) = g[h(u)]$ is a convex function.